GABRIELLA'S MATH LAB · DECIMALS WORKBOOK

SECTION 2

Reading, Writing, and Comparing Decimals

Say a decimal, write it three ways, and decide which decimal is greater.

SKILL
One new idea at a time.

SHOW HOW
Watch a problem solved step by step.

TRY ONE
Finish a problem with help.

YOUR TURN
Solve problems on your own.

2.1	Reading Decimals in Words	☐ done	2
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Name _____ Started _____

Georgia K-12 Mathematics Standards 5.NR.4.1 and 5.NR.4.2. Builds on Section 1: Decimal Place Value.

2.1 Reading Decimals in Words

Goal: I can read and write a decimal in words.

SHOW HOW

Write 4.735 in words.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
4	.	7	3	5

1. Read the **whole-number part**: **4** → “four”.
2. Say **“and”** for the decimal point.
3. Read the digits after the point as one number: 735 → “seven hundred thirty-five”.
4. Say the place of the **last digit**. The 5 is in the **thousandths** place → “thousandths”.

four and seven hundred thirty-five thousandths

Why it works: 0.735 is 735 thousandths, so saying the last place names the size of every piece. Use “and” **only** for the decimal point.

LET'S TRY ONE

Write 12.05 in words.

Whole-number part: _____ then say “and”.

Digits after the point: _____ Last digit's place: _____

In words: _____

YOUR TURN — SKILL 2.1 · WRITE EACH DECIMAL IN WORDS

1 **0.6**

Answer _____

2 **0.38**

Answer _____

3 **5.2**

Answer _____

4 **0.07**

Answer _____

5 **3.45**

Answer _____

6 **0.209**

Answer _____

7 **16.08**

Answer _____

8 **7.125** (a cheetah's time in seconds)

Answer _____

2.2 Writing Decimals from Words

Goal: I can write the standard form of a decimal given in words.

SHOW HOW

Write “six and eight hundredths” as a decimal.

Ones 1		Tenths 0.1	Hundredths 0.01
6	.	0	8

1. The word before **“and”** is the whole number: **6**. Write **6.**
2. The last word is **hundredths**, so there are **2** places after the point.
3. “Eight” must end in the hundredths place: **6._8**
4. Fill the empty tenths place with a **0**: **6.08** *not 6.8!*

Why it works: the place word tells you how many digits go after the point: tenths = 1, hundredths = 2, thousandths = 3.

LET'S TRY ONE

Write “three hundred twelve thousandths” as a decimal.

Is there an “and”? _____ So the whole-number part is _____

Last word: _____ → _____ places after the point.

Decimal: _____

YOUR TURN — SKILL 2.2 · WRITE EACH NUMBER AS A DECIMAL

1 **nine tenths**

Decimal _____

2 **forty-five hundredths**

Decimal _____

3 **two and seven tenths**

Decimal _____

4 **six hundredths**

Decimal _____

5 **fourteen and three tenths**

Decimal _____

6 **five hundred eight thousandths**

Decimal _____

7 **nine and four hundredths**

Decimal _____

8 A raindrop weighs **thirteen thousandths** of a gram. Write the decimal.

Decimal _____

2.3 Standard, Word, and Expanded Form

Goal: I can change a decimal from one form to another.

SHOW HOW

Write 2.406 in word form and expanded form.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
2	.	4	0	6
2 ones **2**		4 tenths **0.4**	0 hundredths **0**	6 thousandths **0.006**

Standard form	**2.406**	the short way, digits in their places
Word form	**two and four hundred six thousandths**	“and”, then read 406, then the last place
Expanded form	**2 + 0.4 + 0.006**	add the value of each digit; skip the 0

Going backward? Write the standard form first, then change it: **2 + 0.4 + 0.006 = 2.406** → “two and four hundred six thousandths”. All three forms name the same number.

LET'S TRY ONE

Write 5.38 in word form and expanded form.

Word form: _____

Expanded form: _____ + _____ + _____

YOUR TURN — SKILL 2.3 · CHANGE THE FORM

1 Write **“seventy-four hundredths”** in expanded form.

Expanded _____

2 Write **“three and six tenths”** in expanded form.

Expanded _____

3 Write **“eight and five hundredths”** in expanded form.

Expanded _____

4 Write **“three hundred seventeen thousandths”** in expanded form.

Expanded _____

5 Write **0.3 + 0.04** in word form.

Answer _____

6 Write **6 + 0.9** in word form.

Answer _____

7 Write **1 + 0.2 + 0.007** in word form.

Answer _____

8 Write **20 + 0.05** in word form.

Answer _____

2.4 Comparing Decimals with >, <, =

Goal: I can compare two decimals using place value.

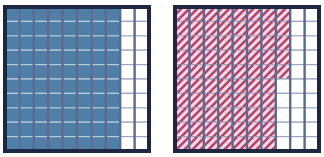
SHOW HOW

Compare 0.8 and 0.75. Write >, <, or =.

Ones 1		Tenths 0.1	Hundredths 0.01
0	.	8	0

Ones 1		Tenths 0.1	Hundredths 0.01
0	.	7	5

1. **Line up** the decimal points.
2. Give both the same number of places: 0.8 = 0.80.
3. Compare from the **left**. Ones: 0 = 0. Tenths: **8 > 7**. Stop: the first different place decides.
4. **0.8 > 0.75**



0.80 0.75

Why it works: 0.80 is 80 hundredths and 0.75 is 75 hundredths. A tenth is worth 10 hundredths, so 8 tenths beat 7 tenths no matter what comes after. More digits does **not** mean bigger. If every digit matches after adding zeros, they are **equal**: 0.7 = 0.70.

LET'S TRY ONE

Compare 3.42 and 3.408.

Same number of places: 3.42 = _____ and 3.408

First place that is different: _____ _____ vs _____

3.42 ☐ **3.408**

YOUR TURN — SKILL 2.4 · WRITE >, <, OR =

1 0.4 ☐ 0.6

2 0.7 ☐ 0.70

3 0.35 ☐ 0.4

4 2.08 ☐ 2.8

5 0.519 ☐ 0.52

6 1.3 ☐ 0.98

7 6.250 ☐ 6.25

8 Jordan ran **0.099** mile. Ava ran **0.1** mile.

0.099 ☐ 0.1

2.5 Comparing on a Number Line

Goal: I can read decimals on a number line and compare them.

SHOW HOW

Points A and B are on the number line. Which is greater?



1. Find the size of **one jump**: from 0.3 to 0.4 in 10 jumps, so each jump is **0.01**. (From 0.6 to 0.61, each jump is 0.001.)
2. Read each point: A is 2 jumps past 0.3 → **0.32**. B is 7 jumps past 0.3 → **0.37**.
3. On a number line, numbers get **greater to the right**. B is to the right of A.
4. **0.37 > 0.32**

Why it works: each jump to the right adds the same amount, so the point farther right is always the greater number.

LET'S TRY ONE

Read points A and B. Then compare.



Each jump is _____ A = _____ B = _____

A ☐ **B**

YOUR TURN — SKILL 2.5 · READ A AND B, THEN WRITE >, <, OR =

1



A = _____ B = _____ A ☐ B

2



A = _____ B = _____ A ☐ B

3



A = _____ B = _____ A ☐ B

4



A = _____ B = _____ A ☐ B

5



A = _____ B = _____ A ☐ B

6



A = _____ B = _____ A ☐ B

7



A = _____ B = _____ A ☐ B

8



A = _____ B = _____ A ☐ B

2.6 Ordering from Least to Greatest

Goal: I can put decimals in order from least to greatest.

SHOW HOW

Order 0.5, 0.45, 0.405 from least to greatest.

0.500

0.450

0.405

1. **Line up** the decimal points and write zeros so all have 3 places.
2. Compare the **tenths**: 5 is biggest, so **0.5** is the greatest.
3. 0.450 and 0.405 tie in tenths (4). Compare **hundredths**: 0 < 5, so **0.405** is the least.
4. Least to greatest: **0.405, 0.45, 0.5**

Why it works: ordering is comparing again and again. Start at the biggest place; only go to the next place when digits tie.

LET'S TRY ONE

Order 2.3, 2.03, 2.33 from least to greatest.

2.3 = 2._____

2.03 = 2.03

2.33 = 2.33

Ones all tie (2). Least tenths digit: _____

So the least number is _____

Least to greatest: _____, _____, _____

YOUR TURN — SKILL 2.6 · ORDER FROM LEAST TO GREATEST

1 **0.6, 0.2, 0.9**

Least → greatest _____

2 **0.34, 0.43, 0.3**

Least → greatest _____

3 **1.5, 1.05, 1.55**

Least → greatest _____

4 **0.7, 0.07, 0.77**

Least → greatest _____

5 **3.21, 3.12, 3.2, 3.1**

Least → greatest _____

6 **0.809, 0.89, 0.8**

Least → greatest _____

7 **4.06, 4.6, 4.006, 4.066**

Least → greatest _____

8 Three pencils are **2.45** in, **2.5** in, and **2.405** in long. Order from shortest to longest.

Least → greatest _____

2.7 Ordering from Greatest to Least

Goal: I can put decimals in order from greatest to least.

SHOW HOW

Order 1.2, 1.02, 1.202 from greatest to least.

1.200

1.020

1.202

1. Line up the points; write zeros so all have 3 places. The ones all tie (1).
2. **Tenths**: 1.02 has 0 tenths, so it is the **least**.
3. 1.200 and 1.202 tie in tenths and hundredths. **Thousandths**: 2 > 0, so **1.202** is the greatest.
4. Greatest to least: **1.202, 1.2, 1.02**

Why it works: it is the same comparing as 2.6. Just write the list starting with the **greatest** number. Check: the first number should be the biggest.

LET'S TRY ONE

Order 0.7, 0.67, 0.076 from greatest to least.

0.7 = 0._____

0.67 = 0._____

0.076 = 0.076

Greatest tenths digit: _____

So the greatest number is _____

Greatest to least: _____, _____, _____

YOUR TURN — SKILL 2.7 · ORDER FROM GREATEST TO LEAST

1 **0.3, 0.8, 0.5**

Greatest → least _____

2 **0.62, 0.26, 0.6**

Greatest → least _____

3 **5.4, 5.04, 5.44**

Greatest → least _____

4 **0.09, 0.9, 0.19**

Greatest → least _____

5 **7.35, 7.53, 7.5, 7.3**

Greatest → least _____

6 **0.415, 0.45, 0.4**

Greatest → least _____

7 **9.1, 9.01, 9.11, 9.011**

Greatest → least _____

8 Three frogs jumped **1.25** m, **1.3** m, and **1.205** m. Order from the longest jump to the shortest.

Greatest → least _____

Section 2 Review

Mixed practice from Skills 2.1 to 2.7. No hints: show what you know.

YOUR TURN **REVIEW · PROBLEMS 1–6**

1 Write **8.36** in words.

Answer _____

2 Write **"forty-two thousandths"** as a decimal.

Decimal _____

3 Write **"four and nine hundred three thousandths"** in expanded form.

Expanded _____

4 **0.6** ☐ **0.59**

5 **3.050** ☐ **3.05**

6 **0.204** ☐ **0.24**

YOUR TURN — REVIEW · PROBLEMS 7–10

7 Read A and B. Then compare.



A = _____ B = _____ A ☐ B

8 Order from least to greatest:
0.57, 0.5, 0.507

Least → greatest _____

9 Order from greatest to least:
12.4, 12.04, 12.44

Greatest → least _____

10 Leo ran in **4.75** seconds. Mia ran in **4.8** seconds. Who took **longer**? Explain with place value.

Who _____

HOW DID IT GO?

Circle the skills to practice again, then redo two problems from that skill's Your Turn page. **2.1** **2.2** **2.3** **2.4** **2.5** **2.6** **2.7**

Parent Answer Key

Equal values are correct too (0.7 = 0.70). Accept word form with or without hyphens. "Try" is the Let's Try One problem.

2.1 Reading Decimals in Words

Try **twelve and five hundredths** 12 wholes, then "and" for the point; read 5, then say the place of the last digit: hundredths.

1. **six tenths** Read 6, then say the place of the last digit: tenths.
2. **thirty-eight hundredths** Read 38, then say the place of the last digit: hundredths.
3. **five and two tenths** 5 wholes, then "and" for the point; read 2, then say the place of the last digit: tenths.
4. **seven hundredths** Read 7, then say the place of the last digit: hundredths.
5. **three and forty-five hundredths** 3 wholes, then "and" for the point; read 45, then say the place of the last digit: hundredths.
6. **two hundred nine thousandths** Read 209, then say the place of the last digit: thousandths.
7. **sixteen and eight hundredths** 16 wholes, then "and" for the point; read 8, then say the place of the last digit: hundredths.
8. **seven and one hundred twenty-five thousandths** 7 wholes, then "and" for the point; read 125, then say the place of the last digit: thousandths.

2.2 Writing Decimals from Words

Try **0.312** No "and", so the whole-number part is 0. Thousandths = 3 places after the point: 312 fills them. 0.312.

1. **0.9** No "and", so 0 wholes. Tenths = 1 place after the point: 9 fills them → 0.9.
2. **0.45** No "and", so 0 wholes. Hundredths = 2 places after the point: 45 fills them → 0.45.
3. **2.7** Before "and": 2. Tenths = 1 place after the point: 7 fills them → 2.7.
4. **0.06** No "and", so 0 wholes. Hundredths = 2 places after the point; 6 has 1 digit, so write a 0 first → 0.06.
5. **14.3** Before "and": 14. Tenths = 1 place after the point: 3 fills them → 14.3.
6. **0.508** No "and", so 0 wholes. Thousandths = 3 places after the point: 508 fills them → 0.508.
7. **9.04** Before "and": 9. Hundredths = 2 places after the point; 4 has 1 digit, so write a 0 first → 9.04.
8. **0.013** No "and", so 0 wholes. Thousandths = 3 places after the point; 13 has 2 digits, so write a 0 first → 0.013.

2.3 Standard, Word, and Expanded Form

Try **5 + 0.3 + 0.08** Word form: five and thirty-eight hundredths. Expanded form: 5 + 0.3 + 0.08.

1. **0.7 + 0.04** 7 tenths, 4 hundredths.
2. **3 + 0.6** 3 ones, 6 tenths.
3. **8 + 0.05** 8 ones, 5 hundredths (the 0 in the tenths place adds nothing).
4. **0.3 + 0.01 + 0.007** 3 tenths, 1 hundredth, 7 thousandths.
5. **thirty-four hundredths** 0.3 + 0.04 = 0.34; read 34, then say the place of the last digit: hundredths.
6. **six and nine tenths** 6 + 0.9 = 6.9; 6 wholes, then "and" for the point; read 9, then say the place of the last digit: tenths.
7. **one and two hundred seven thousandths** 1 + 0.2 + 0.007 = 1.207; 1 whole, then "and" for the point; read 207, then say the place of the last digit: thousandths.
8. **twenty and five hundredths** 20 + 0.05 = 20.05; 20 wholes, then "and" for the point; read 5, then say the place of the last digit: hundredths.

2.4 Comparing Decimals with >, <, =

Try **>** Write 3.420 and 3.408. First different place: hundredths, 2 > 0. So 3.42 > 3.408.

1 **<** Write 0.4 and 0.6. First different place: tenths, 4 < 6. So 0.4 < 0.6.

2 **=** 0.7 = 0.70: written with the same number of places, the digits are all the same.

3 **<** Write 0.35 and 0.40. First different place: tenths, 3 < 4. So 0.35 < 0.4.

4 **<** Write 2.08 and 2.80. First different place: tenths, 0 < 8. So 2.08 < 2.8.

5 **<** Write 0.519 and 0.520. First different place: hundredths, 1 < 2. So 0.519 < 0.52.

6 **>** Write 1.30 and 0.98. First different place: ones, 1 > 0. So 1.3 > 0.98.

7 **=** 6.250 = 6.25: written with the same number of places, the digits are all the same.

8 **<** Write 0.099 and 0.100. First different place: tenths, 0 < 1. So 0.099 < 0.1.

2.5 Comparing on a Number Line

Try **>** Each jump is 0.1. A is 6 jumps past 1, so A = 1.6. B is 3 jumps past 1, so B = 1.3. A is farther right: 1.6 > 1.3.

1 **<** Each jump is 0.1. A = 0.2, B = 0.7. B is farther right, so A < B.

2 **>** Each jump is 0.1. A = 0.9, B = 0.4. A is farther right, so A > B.

3 **<** Each jump is 0.1. A = 2.5, B = 2.8. B is farther right, so A < B.

4 **>** Each jump is 0.01. A = 0.46, B = 0.41. A is farther right, so A > B.

5 **<** Each jump is 0.01. A = 0.73, B = 0.79. B is farther right, so A < B.

6 **>** Each jump is 0.01. A = 0.08, B = 0.03. A is farther right, so A > B.

7 **>** Each jump is 0.01. A = 5.25, B = 5.2. A is farther right, so A > B.

8 **<** Each jump is 0.001. A = 0.604, B = 0.609. B is farther right, so A < B.

2.6 Ordering from Least to Greatest

Try **2.03, 2.3, 2.33** With 2 decimal places: 2.03 < 2.30 < 2.33.

1 **0.2, 0.6, 0.9** With 1 decimal place: 0.2 < 0.6 < 0.9.

2 **0.3, 0.34, 0.43** With 2 decimal places: 0.30 < 0.34 < 0.43.

3 **1.05, 1.5, 1.55** With 2 decimal places: 1.05 < 1.50 < 1.55.

4 **0.07, 0.7, 0.77** With 2 decimal places: 0.07 < 0.70 < 0.77.

5 **3.1, 3.12, 3.2, 3.21** With 2 decimal places: 3.10 < 3.12 < 3.20 < 3.21.

6 **0.8, 0.809, 0.89** With 3 decimal places: 0.800 < 0.809 < 0.890.

7 **4.006, 4.06, 4.066, 4.6** With 3 decimal places: 4.006 < 4.060 < 4.066 < 4.600.

8 **2.405, 2.45, 2.5** With 3 decimal places: 2.405 < 2.450 < 2.500.

2.7 Ordering from Greatest to Least

Try **0.7, 0.67, 0.076** With 3 decimal places: 0.700 > 0.670 > 0.076.

1 **0.8, 0.5, 0.3** With 1 decimal place: 0.8 > 0.5 > 0.3.

2 **0.62, 0.6, 0.26** With 2 decimal places: 0.62 > 0.60 > 0.26.

3 **5.44, 5.4, 5.04** With 2 decimal places: 5.44 > 5.40 > 5.04.

4 **0.9, 0.19, 0.09** With 2 decimal places: 0.90 > 0.19 > 0.09.

5 **7.53, 7.5, 7.35, 7.3** With 2 decimal places: 7.53 > 7.50 > 7.35 > 7.30.

6 **0.45, 0.415, 0.4** With 3 decimal places: 0.450 > 0.415 > 0.400.

7 **9.11, 9.1, 9.011, 9.01** With 3 decimal places: 9.110 > 9.100 > 9.011 > 9.010.

8 **1.3, 1.25, 1.205** With 3 decimal places: 1.300 > 1.250 > 1.205.

Review Section 2 Review

1 **eight and thirty-six hundredths** 8 wholes, then "and" for the point; read 36, then say the place of the last digit: hundredths.

2 **0.042** No "and", so 0 wholes. Thousandths = 3 places after the point; 42 has 2 digits, so write a 0 first → 0.042.

3 **4 + 0.9 + 0.003** 4 ones, 9 tenths, 3 thousandths (the 0 in the hundredths place adds nothing).

4 **>** Write 0.60 and 0.59. First different place: tenths, 6 > 5. So 0.6 > 0.59.

5 **=** 3.050 = 3.05: written with the same number of places, the digits are all the same.

6 **<** Write 0.204 and 0.240. First different place: hundredths, 0 < 4. So 0.204 < 0.24.

7 **<** Each jump is 0.01. A = 0.75, B = 0.8. B is farther right, so A < B.

8 **0.5, 0.507, 0.57** With 3 decimal places: 0.500 < 0.507 < 0.570.

9 **12.44, 12.4, 12.04** With 2 decimal places: 12.44 > 12.40 > 12.04.

10 **Mia** 4.8 = 4.80. Compare 4.80 and 4.75: ones tie (4); tenths 8 > 7. So 4.8 > 4.75: Mia took longer. (More digits does not mean more time.)